

Umaid Ali

Lahore, Punjab, Pakistan

aliumaid64488@gmail.com | +92 314 735 7980

GitHub: Umaid-Ali-804 | LinkedIn: umaid-ali | Medium: @aliumaid64488

Professional Summary

AI Engineer pursuing an MS in Data Science with experience designing and deploying intelligent AI systems using **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **Machine Learning**, **Deep Learning**, and **Multi-Agent AI**. Skilled in building scalable AI applications with **Python**, **PyTorch**, **LangChain**, **LangGraph**, **FastAPI**, **Docker**, and **vector databases**. Experienced in developing production-ready AI pipelines, model evaluation, and LLM safety solutions with a strong interest in AI research and real-world deployment.

Experience

Associate AI Engineer | Graduate Research Assistant

Jan 2025 – Present

Information Technology University (ITU), Lahore

- Designed and developed production-ready AI applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, FastAPI, and vector databases for intelligent document understanding, semantic search, and domain-specific question answering.
- Engineered scalable AI pipelines encompassing document preprocessing, semantic chunking, embedding generation, hybrid retrieval, reranking, prompt engineering, agentic workflows, and RESTful API integration to deliver reliable and efficient AI solutions.
- Conducting research on trustworthy Generative AI by developing hallucination detection frameworks using Natural Language Inference (NLI), neural-symbolic learning, uncertainty estimation, and biomedical knowledge graphs, while translating research innovations into deployable AI systems.

Projects

ClinHallu: Medical Hallucination Detection System (Master's Thesis) | Python, PyTorch, PEFT, DeBERTa

- Developed a novel neural-symbolic medical hallucination detection framework integrating transformer-based encoder models, biomedical knowledge graphs, and Natural Language Inference (NLI) for trustworthy evaluation of medical Large Language Models (LLMs).
- Constructed the **CDRC (Clinical Dual-Role Corpus)**, a large-scale medical dataset containing over **120,000** annotated samples, and trained the proposed framework using this dataset before evaluating it on the **MedHallu** benchmark.
- Achieved a **Binary F1-score of 85%** and **ROC-AUC of 91%** on the MedHallu benchmark, outperforming existing computationally expensive approaches while introducing a dedicated hallucination detection framework for the medical domain.

AI-Powered Insurance Claims Processing Assistant | Python, PyTorch, YOLOv8, LangGraph, FastAPI, Docker

- Built an end-to-end AI platform automating insurance claim processing using computer vision, RAG, and LLMs. Developed multimodal pipelines for vehicle damage detection, OCR, fraud prediction, and AI-based claim verification.
- Implemented a LangGraph multi-agent workflow and deployed scalable REST APIs using FastAPI and Docker.

AI-Assisted Medical Coding System | Python, LLMs, PostgreSQL, SQL, RAG, pgvector

- Developed a multi-agent medical coding system integrating LLMs, SQL-based retrieval, RAG, and deterministic rule validation for ICD-10-CM, CPT, and HCPCS coding.
- Designed a seven-stage pipeline for clinical extraction, candidate retrieval, code selection, rule validation, guideline-based verification, and confidence scoring with evidence traceability and human-in-the-loop review.

Technical Skills

Machine Learning: Python, SQL, Scikit-learn, XGBoost, PyTorch, TensorFlow

Generative AI: LangChain, LangGraph, CrewAI, AutoGen, Prompt Engineering, RAG, AI Agents, n8n

Large Language Models: OpenAI API, Anthropic API, Gemini API, Hugging Face Transformers

Computer Vision: YOLOv8, OpenCV, OCR

Vector Databases: FAISS, Pinecone, Qdrant, pgvector

MLOps & Deployment: FastAPI, Docker, MLflow, Git, GitHub Actions, CI/CD, AWS (EC2, S3)

Evaluation & AI Safety: RAGAS, TruLens, SHAP, LLM Evaluation, Hallucination Detection

Education

MS in Data Science

Jun 2024 – Present

Information Technology University (ITU), Lahore

BS Engineering

2019 – 2023

University of Engineering and Technology (UET), Lahore

Certifications

- Deep Learning Specialization – Coursera (Andrew Ng), 2023